

Convergence Rates

Adaptive Finite Element Methods for Elliptic PDEs — Lecture 9

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Optimal convergence rates

- **Goal:** We complement the *contraction* analysis of Lecture 8 with **optimal convergence rates**:

the error decay matches the rate of the *best possible mesh* with the same number of degrees of freedom.

- **Two-stage theory (mirroring the contraction theory):**

- **Vanishing oscillation** (piecewise-constant f, A): oscillation vanishes $\text{osc}_{\mathcal{T}}(U) = 0$, the error and the estimator are equivalent, whence *energy error* alone measures the rate.
- **General data:** the natural quantity to measure performance of AFEM is the *total error*

$$E_{\mathcal{T}}(U) = \|u - U\|_{\Omega} + \text{osc}_{\mathcal{T}}(U).$$

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- **General data:** the natural quantity to measure performance of AFEM is the *total error*

$$E_{\mathcal{T}}(U) = \|u - U\|_{\Omega} + \text{osc}_{\mathcal{T}}(U).$$

- **The punchline:**

AFEM *learns* the unknown regularity $\mathcal{A}_s$ of the solution from the computed a posteriori indicators, and builds a quasi-optimal mesh with **linear complexity** — with *no* a priori knowledge of the singularity.

Roadmap

- 1 Setup and basic assumptions
- 2 Vanishing oscillation: piecewise-constant data
- 3 General data: the total error
- 4 Quasi-optimal rates for general data
- 5 Synthesis
- 6 A new paradigm of AFEM

Setup and basic assumptions

Three basic assumptions

Assumption 1 (marking parameter)

There is a threshold $0 < \theta_* \leq 1$ such that the Dörfler parameter satisfies $\theta \in (0, \theta_*)$:

$$\mathcal{E}_{\mathcal{T}}^2(U, \mathcal{M}) = \sum_{T \in \mathcal{M}} \mathcal{E}_{\mathcal{T}}^2(U, T) \geq \theta^2 \sum_{T \in \mathcal{T}} \mathcal{E}_{\mathcal{T}}^2(U, T) = \theta^2 \mathcal{E}_{\mathcal{T}}^2(U).$$

Assumption 2 (minimal cardinality)

MARK selects a set $\mathcal{M} \subset \mathcal{T}$ of marked elements for refinement with *minimal* cardinality.

Assumption 3 (initial labeling)

The initial mesh $\mathcal{T}_0$ carries a compatible labeling (so that the statement about complexity of bisection holds for $d = 2$, or its multidimensional counterpart for $d > 2$):

$$\#\mathcal{T} - \#\mathcal{T}_0 \leq \Lambda_0 \sum_{j=0}^{k-1} \#\mathcal{M}_j.$$

Interpreting the assumptions

- Threshold $\theta_* < 1$ (prudent marking):** If the reliability/efficiency constants C_1, C_2 are disparate (i.e. ratio C_2/C_1 far from 1), one must be cautious — hence $\theta_* < 1$.
- Minimal $\mathcal{M}$:** By error equidistribution and the local lower bound $C_2 \mathcal{E}_{\mathcal{T}}(U, T) \lesssim \|u - U\|_{\omega_T}$, provided $\text{osc}_{\mathcal{T}}(U) = 0$, it is natural to mark the elements with **largest indicators $\mathcal{E}_{\mathcal{T}}(U, T)$** . This yields a minimal set $\mathcal{M}$ and links AFEM to optimal meshes.
- Initial labeling:** This guarantees the linear complexity of bisection: REFINES produces conforming refinements $\mathcal{T}_k \geq \mathcal{T}_0$ with cardinality

$$\#\mathcal{T}_k - \#\mathcal{T}_0 \leq \Lambda_0 \sum_{j=0}^{k-1} \#\mathcal{M}_j$$

governed by the cumulative number of marked elements. Therefore, to count $\#\mathcal{T}_k$ we just **have to estimate $\sum_{j=0}^{k-1} \#\mathcal{M}_j$** .

Vanishing oscillation: piecewise-constant data

Setting: vanishing oscillation

- Piecewise-constant data:** Throughout this stage, data (f, A) are piecewise constant on $\mathcal{T}_0$, so $\text{osc}_k(U_k) \equiv 0$ on every refinement $\mathcal{T}_k \geq \mathcal{T}_0$, and the estimator $\mathcal{E}_k = \mathcal{E}_{\mathcal{T}_k}(U_k)$ is two-sided equivalent to the energy error $e_k := \|u - U_k\|_\Omega$:

$$C_2 \mathcal{E}_k^2 \leq e_k^2 \leq C_1 \mathcal{E}_k^2.$$

- Performance of AFEM:** The achievable convergence rate is then measured *purely* by the energy error e_k . This is the quantity to monitor to assess optimality.

Approximation classes

- **Best N -element error:** Let $\mathbb{T}_N$ be all conforming bisection refinements $\mathcal{T}$ of $\mathcal{T}_0$ with at most N extra elements, and define

$$\sigma_N(u) = \inf_{\mathcal{T} \in \mathbb{T}_N} \inf_{V \in \mathbb{V}_{\mathcal{T}}} \|u - V\|_{\Omega}.$$

- **The class $\mathcal{A}_s$:** A function $u \in \mathcal{A}_s$ if and only if

$$|u|_s := \sup_{N \in \mathbb{N}} \left(N^s \sigma_N(u) \right) < \infty, \quad \text{i.e.} \quad \sigma_N(u) \leq |u|_s N^{-s}.$$

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- **Interpretation:**

- $\mathcal{A}_s$ is the class of all functions that can be approximated with rate N^{-s} .
- Finding a near-best mesh has, in general, **exponential** complexity in N .
- AFEM achieves optimal rate with **linear** complexity (related to Besov regularity; a characterization of $\mathcal{A}_s$ is known.)

Membership in $\mathcal{A}_{1/2}$

Lemma 1 (membership in $\mathcal{A}_{1/2}$ for piecewise-constant (A, f))

Let $d = 2$ and $u \in H_0^1(\Omega) \cap W_p^2(\Omega)$, $p > 1$, solve $-\operatorname{div}(A\nabla u) = f$ on a polygon Ω (reentrant corners allowed). If Assumption 3 (initial labeling) hold, then $u \in \mathcal{A}_{1/2}$ with $|u|_{1/2} \lesssim |u|_{W_p^2(\Omega)}$.

- **Proof:** Apply the optimal greedy-approximation theorem (Lecture 4) to deduce the existence of $\mathcal{T} \in \mathbb{T}_N$ so that

$$\|u - I_{\mathcal{T}} u\|_{\Omega} \lesssim |u|_{W_p^2(\Omega)} N^{-1/2}.$$

The definitions of σ_N and $\mathcal{A}_s$ give the assertion. □

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- **Examples:** For $d = 2$ the L-shaped domain and the Kellogg problem belong to $\mathcal{A}_{1/2}$.
- **Key point:**

AFEM realizes the rate $N^{-1/2}$ — i.e. $s = \frac{1}{2}$ — without explicit knowledge of the singularity of $u \in \mathcal{A}_s$.

A useful reformulation of $\mathcal{A}_s$

• Exercise (ε -characterization of $\mathcal{A}_s$):

$u \in \mathcal{A}_s$ if and only if for every $\varepsilon > 0$ there is a conforming refinement $\mathcal{T}_\varepsilon \geq \mathcal{T}_0$ and $V_\varepsilon \in \mathbb{V}_{\mathcal{T}_\varepsilon}$ with

$$\|u - V_\varepsilon\|_\Omega \leq \varepsilon \quad \text{and} \quad \#\mathcal{T}_\varepsilon - \#\mathcal{T}_0 \lesssim |u|_s^{1/s} \varepsilon^{-1/s}.$$

- **Interpretation:** Belonging to class $\mathcal{A}_s$ means that to achieve an error tolerance ε we need an order $\varepsilon^{-1/s}$ elements (or degrees of freedom or computational budget).

Optimal marking — Stevenson's insight

- **Heuristic insight for the Laplacian (piecewise-constant f):**

*Any marking strategy that reduces the energy error relative to its current value must capture a substantial portion of $\mathcal{E}_{\mathcal{T}}(U)$ — hence it can be related to **Dörfler marking**.*

- **Consequence:** This lets us compare AFEM meshes with optimal ones and conclude **quasi-optimal error decay** for piecewise constant data (A, f) . For general data we need to monitor the total error instead. We do this later.

Dörfler marking is necessary (vanishing oscillation)

Lemma 2 (Dörfler marking: vanishing oscillation)

Let Assumption 1 (marking parameter) hold with threshold $\theta_* = \sqrt{C_2/C_1}$ and set $\mu := 1 - \theta^2/\theta_*^2 > 0$ with $\theta \in (0, \theta_*)$. If a conforming refinement $\mathcal{T}_* \geq \mathcal{T}_k$ with Galerkin solution U_* satisfies

$$\|u - U_*\|_{\Omega}^2 \leq \mu \|u - U_k\|_{\Omega}^2,$$

then the discrete neighborhood $\mathcal{R}^+$ of the refined set $\mathcal{R} = \mathcal{T}_k \setminus \mathcal{T}_*$ satisfies Dörfler marking

$$\mathcal{E}_k(U_k, \mathcal{R}^+) \geq \theta \mathcal{E}_k(U_k, \mathcal{T}_k).$$

- **Interpretation:** enough energy reduction relative to the current error implies that the refined elements carry a fixed fraction of the current global estimator.

Proof (energy reduction $\Rightarrow$ Dörfler marking)

- Pythagoras equality:

$$\|u - U_k\|_\Omega^2 - \|u - U_*\|_\Omega^2 = \|U_k - U_*\|_\Omega^2.$$

- Lower bound: $C_2 \mathcal{E}_k^2 \leq e_k^2$ yields

$$(1 - \mu)C_2 \mathcal{E}_k^2 \leq (1 - \mu)e_k^2 = e_k^2 - \mu e_k^2 \leq \|u - U_k\|_\Omega^2 - \|u - U_*\|_\Omega^2 = \|U_k - U_*\|_\Omega^2.$$

- Localized upper bound: $\|U_k - U_*\|_\Omega^2 \leq C_1 \mathcal{E}_k^2(U_k, \mathcal{R}^+)$ gives

$$(1 - \mu)C_2 \mathcal{E}_k^2 \leq C_1 \mathcal{E}_k^2(U_k, \mathcal{R}^+) \Rightarrow \mathcal{E}_k(U_k, \mathcal{R}^+) \geq \theta \mathcal{E}_k, \quad \theta^2 = (1 - \mu)\frac{C_2}{C_1} < \theta_*^2 = \frac{C_2}{C_1}. \quad \square$$

Mesh overlay

- **Overlay of two meshes:**

Given conforming refinements $\mathcal{T}_1, \mathcal{T}_2$ of $\mathcal{T}_0$, the overlay $\mathcal{T} = \mathcal{T}_1 \oplus \mathcal{T}_2$ is the smallest common conforming refinement of $\mathcal{T}_1, \mathcal{T}_2$, and $\mathcal{T}$ satisfies

$$\#\mathcal{T} \leq \#\mathcal{T}_1 + \#\mathcal{T}_2 - \#\mathcal{T}_0.$$

- **Role of overlay:** The overlay allows us to compare the k -th AFEM mesh $\mathcal{T}_k$ with a near-best mesh $\mathcal{T}_\varepsilon$, which may be completely unrelated to $\mathcal{T}_k$. Therefore, it is the link between actual finite element meshes and optimal ones.

Cardinality of the marked set $\mathcal{M}_k$

Lemma 3 (cardinality of $\mathcal{M}_k$: vanishing oscillation)

Let Assumptions 1 (marking parameter) with threshold $\theta_* = \sqrt{C_2/C_1}$, 2 (minimal cardinality) and 3 (initial labeling) hold. If $u \in \mathcal{A}_s$ with $0 < s \leq \frac{1}{2}$, then

$$\#\mathcal{M}_k \lesssim |u|_s^{1/s} \|u - U_k\|_{\Omega}^{-1/s} \quad \forall k \geq 1.$$

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• **Proof. Optimal mesh and overlay:** Take $\varepsilon^2 = \mu e_k^2$ in the ε -characterization: get $\mathcal{T}_\varepsilon$ with

$$\|u - U_\varepsilon\|_\Omega \leq \varepsilon, \quad \#\mathcal{T}_\varepsilon - \#\mathcal{T}_0 \lesssim |u|_s^{1/s} \varepsilon^{-1/s}.$$

Letting $\mathcal{T}_* = \mathcal{T}_k \oplus \mathcal{T}_\varepsilon$ be the overlay of $\mathcal{T}_k$ and $\mathcal{T}_\varepsilon$, the Galerkin solution $U_* \in \mathbb{V}_*$ satisfies

$$\|u - U_*\|_\Omega^2 \leq \|u - U_\varepsilon\|_\Omega^2 \leq \varepsilon^2 = \mu e_k^2.$$

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- **Error reduction:** Lemma 2 implies that the discrete neighborhood $\mathcal{R}^+$ of the refined set $\mathcal{R} = \mathcal{T}_k - \mathcal{T}_*$ satisfies Dörfler marking.

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- **Error reduction:** Lemma 2 implies that the discrete neighborhood $\mathcal{R}^+$ of the refined set $\mathcal{R} = \mathcal{T}_k - \mathcal{T}_*$ satisfies Dörfler marking.
- **Minimal cardinality:** Using that $\mathcal{M}_k$ is minimal for Dörfler marking and overlay property yields

$$\#\mathcal{M}_k \leq \#\mathcal{R}^+ \lesssim \#\mathcal{R} = \#\mathcal{T}_* - \#\mathcal{T}_k \leq \#\mathcal{T}_\varepsilon - \#\mathcal{T}_0 \lesssim |u|_s^{1/s} \varepsilon^{-1/s}. \quad \square$$

Quasi-optimal convergence rate (piecewise-constant data)

Theorem 1 (quasi-optimal convergence rate)

Let $\text{osc}_k(U_k) = 0$ and Assumptions 1 (marking parameter) with threshold $\theta_* = \sqrt{C_2/C_1}$, 2 (minimal cardinality), and 3 (initial labeling) hold. If the interior-node property is enforced and $u \in \mathcal{A}_s$, $0 < s \leq \frac{1}{2}$, then (for $\#\mathcal{T}_k \geq 2\#\mathcal{T}_0$)

$$\|u - U_k\|_{\Omega} \lesssim |u|_s (\#\mathcal{T}_k)^{-s} \quad \forall k \geq 1.$$

Therefore, AFEM converges with the best possible rate N^{-s} for the class $\mathcal{A}_s$.

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• Proof ingredients:

- complexity of REFINE
- cardinality of $\mathcal{M}_k$
- contraction property

Proof of quasi-optimality

- **Complexity + cardinality:** Complexity of bisection and Lemma 3 (cardinality of $\mathcal{M}_k$) give

$$\frac{1}{2}\#\mathcal{T}_k \leq \#\mathcal{T}_k - \#\mathcal{T}_0 \lesssim \sum_{j=0}^{k-1} \#\mathcal{M}_j \lesssim |u|_s^{1/s} \sum_{j=0}^{k-1} e_j^{-1/s}.$$

- **Contraction:** Since $\text{osc}_k = 0$, the contraction property gives geometric decay $e_k \leq \alpha^{k-j} e_j$ for $k \geq j$ i.e.

$$e_j^{-1/s} \leq \alpha^{(k-j)/s} e_k^{-1/s}.$$

Proof of quasi-optimality

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- **Conclusion:** The geometric series $\sum_j (\alpha^{1/s})^{k-j}$ converges ($\alpha < 1$)

$$\# \mathcal{T}_k \lesssim |u|_s^{1/s} e_k^{-1/s} \iff e_k \lesssim |u|_s (\# \mathcal{T}_k)^{-s}. \quad \square$$

General data: the total error

How to measure performance of AFEM

- **Total error:** For general data (f, A) the oscillation no longer vanishes. What AFEM reduces is the *quasi-error* of Lecture 8, equivalent (by reliability + efficiency) to the **total error** $E_{\mathcal{T}}(U)$

$$E_{\mathcal{T}}(U)^2 := \|u - U\|_{\Omega}^2 + \text{osc}_{\mathcal{T}}^2(U) \simeq \mathcal{E}_{\mathcal{T}}^2(U) \simeq \|u - U\|_{\Omega}^2 + \gamma \mathcal{E}_{\mathcal{T}}^2(U).$$

- **Role of efficiency:** We have not used the lower bound before.

It is *here* — not in the contraction property — that the efficiency (lower) bound is essential: it ties the total error to the estimator.

Quasi-best approximation of AFEM

Lemma 4 (quasi-best approximation / Céa's lemma for the total error)

There is $\Lambda_1 > 0$, depending only on the data (A, f) and shape regularity, such that the Galerkin solution satisfies

$$E_{\mathcal{T}}(U) \leq \Lambda_1 \inf_{V \in \mathbb{V}_{\mathcal{T}}} E_{\mathcal{T}}(V).$$

- Interpretation:** The *total error* $E_{\mathcal{T}}(U)$ is the right quantity to monitor for AFEM because AFEM exhibits a quasi-best approximation property for $E_{\mathcal{T}}(U)$. Therefore, the decay rate of $E_{\mathcal{T}}(U)$ dictates the approximation class for general data.

(Proof is left as an exercise.)

Approximation classes for the triple (u, f, A)

- **Approximation modulus:** Since oscillation $\text{osc}_{\mathcal{T}}(U)$ depends on (f, A) , the notion of approximability must depend on the *triple* (u, f, A) :

$$\sigma_N(u, f, A) = \inf_{\mathcal{T} \in \mathbb{T}_N} \inf_{V \in \mathbb{V}_{\mathcal{T}}} \left(\|u - V\|_{\Omega}^2 + \text{osc}_{\mathcal{T}}^2(V) \right)^{1/2}.$$

It is worth observing that $\sigma_N(u, f, A)$ depends neither on the Galerkin solution $U \in \mathbb{V}_{\mathcal{T}}$ or the estimator $\mathcal{E}_{\mathcal{T}}(U)$, which is not intrinsic but rather dictated by the residual estimator.

- **The class $\mathbb{A}_s$ for (u, f, A) :** The triple $(u, f, A) \in \mathbb{A}_s$ if and only if

$$|(u, f, A)|_s := \sup_{N \in \mathbb{N}} \left(N^s \sigma_N(u, f, A) \right) < \infty.$$

Piecewise constant A : Data oscillation and its class

- **Data oscillation:** For A piecewise constant over $\mathcal{T}_0$, $A\nabla U$ is piecewise constant over $\mathcal{T}$, and the oscillation $\text{osc}_{\mathcal{T}}(U)$ reduces to *data oscillation*

$$\text{osc}_{\mathcal{T}}(f) := \|h(f - \bar{f})\|_{L^2(\Omega)}, \quad \bar{f} = \text{piecewise mean of } f \text{ over } \mathcal{T}.$$

- **The class $\mathcal{B}_t$ for f :**

$$\mathcal{B}_t := \left\{ g \in L^2(\Omega) : |g|_{\mathcal{B}_t} := \sup_{N>0} \left(N^t \inf_{\mathcal{T} \in \mathbb{T}_N} \text{osc}_{\mathcal{T}}(g) \right) < \infty \right\}.$$

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- **Membership in $\mathcal{B}_t$:** For $d = 2$: $f \in W_p^s(\Omega)$, $0 \leq s \leq 1$, gives membership with $t = \frac{1+s}{2} \geq \frac{1}{2}$.

Decay rate of data oscillation

Lemma 5 (decay rate of $\text{osc}_{\mathcal{T}}(f)$)

Let $0 \leq s \leq 1$ with $1 + s - 2/p \geq 0$, $1 \leq p \leq \infty$, and $f \in W_p^s(\Omega)$. If Assumption 3 (initial labeling) is valid and $N \geq 2\#\mathcal{T}_0$, then

$$\inf_{\mathcal{T} \in \mathbb{T}_N} \text{osc}_{\mathcal{T}}(f) \lesssim |f|_{W_p^s(\Omega)} N^{-\frac{1+s}{2}}.$$

- **Proof. Poincaré:** $W_p^s \subset L^2$ since $\text{sob}(W_p^s) - \text{sob}(L^2) \geq 0$. Poincaré (zero mean) gives locally

$$\text{osc}_{\mathcal{T}}(f, T) \lesssim h_T^r |f|_{W_p^s(T)}, \quad r = 2 + s - 2/p.$$

- **Greedy:** Run the GREEDY algorithm of Lecture 4 on $\text{osc}_{\mathcal{T}}(f, T)$ with tolerance δ :

$$\delta \approx (\#\mathcal{T})^{-\frac{1+s}{2} - \frac{1}{2}} |f|_{W_p^s} |\Omega|^{-1/p} \Rightarrow \text{osc}_{\mathcal{T}}^2(f) \leq \delta^2 \#\mathcal{T} \approx (\#\mathcal{T})^{-(1+s)} |f|_{W_p^s}^2 |\Omega|^{-2/p}. \quad \square$$

Equivalence of classes

Lemma 6 (equivalence of classes)

Let A be piecewise constant over $\mathcal{T}_0$ and let Assumption 3 (initial labeling) be enforced. Then

$$(u, f, A) \in \mathbb{A}_s \iff (u, f) \in \mathcal{A}_s \times \mathcal{B}_s, \quad |(u, f, A)|_s \approx |u|_{\mathcal{A}_s} + |f|_{\mathcal{B}_s}.$$

• **Proof:** “ $\Rightarrow$ ” is immediate. For “ $\Leftarrow$ ”, we proceed as follows:

- Pick $\mathcal{T}_1, \mathcal{T}_2 \in \mathbb{T}_N$ with

$$\|u - U\|_{\Omega} \lesssim |u|_{\mathcal{A}_s} N^{-s}, \quad \text{osc}_{\mathcal{T}_2}(f) \lesssim |f|_{\mathcal{B}_s} N^{-s};$$

- The overlay $\mathcal{T} = \mathcal{T}_1 \oplus \mathcal{T}_2 \in \mathbb{T}_{2N}$ satisfies, by **monotonicity of $E_{\mathcal{T}}(U)$** under refinement,

$$\|u - U\| + \text{osc}_{\mathcal{T}}(f) \lesssim (|u|_{\mathcal{A}_s} + |f|_{\mathcal{B}_s}) N^{-s}. \quad \square$$

• **Remark:** Since here $0 < s \leq \frac{1}{2}$ while $t = \frac{1+s}{2} \geq \frac{1}{2}$, the **rate $s \leq \frac{1}{2}$** is dictated **solely by u** .

Membership in $\mathbb{A}_{1/2}$ with piecewise constant A

Corollary 1 (class $\mathbb{A}_{1/2}$ for piecewise-constant A)

Let $u \in W_p^2(\Omega) \cap H_0^1(\Omega)$ ($p > 1$), $f \in L^2(\Omega)$, A is piecewise constant. If Assumption 3 (initial labeling) is enforced, then

$$|(u, f, A)|_{1/2} \lesssim |u|_{W_p^2(\Omega)} + \|f\|_{L^2(\Omega)}.$$

• Proof:

- Greedy approximation (Lecture 4) gives $u \in \mathcal{A}_{1/2}$ with $|u|_{\mathcal{A}_{1/2}} \lesssim |u|_{W_p^2}$
- If $f \in L^2(\Omega)$, then Lemma 5 (decay rate of $\text{osc}_{\mathcal{T}}(f)$) yields $f \in \mathcal{B}_t$ with $t = \frac{1}{2}$.
- Finally, Lemma 6 (equivalence of classes) gives the assertion. □

Variable A : decay rate of $\text{osc}_{\mathcal{T}}(U)$

Lemma 7 (decay rate of $\text{osc}_{\mathcal{T}}(U)$)

Let $A \in W_{\infty}^1(\Omega; \mathcal{T}_0) \cap W_q^{1+s_A}(\Omega; \mathcal{T}_0)$, $s_A - 2/q \geq 0$, and $f \in L^2(\Omega) \cap W_p^{s_f}(\Omega; \mathcal{T}_0)$, $s_f - 2/p + 1 \geq 0$. If Assumption 3 (initial labeling) is valid and $N \geq 2\#\mathcal{T}_0$, then we have with $s = \min\{s_A, s_f\}$

$$\inf_{\mathcal{T} \in \mathbb{T}_N} \text{osc}_{\mathcal{T}}(U) \lesssim \left(|A|_{W_q^{1+s_A}(\Omega; \mathcal{T}_0)} \|f\|_{L^2(\Omega)} + |f|_{W_r^{s_f}(\Omega; \mathcal{T}_0)} \right) N^{-(1+s)/2}.$$

- **Remark:** The *nonlinear* interaction between A and ∇U inside $\text{osc}_{\mathcal{T}}(U)$ is reflected on the product $|A|_{W_q^{1+s_A}(\Omega; \mathcal{T}_0)} \|f\|_{L^2(\Omega)}$. Note that $\|\nabla U\|_{L^2(\Omega)} \lesssim \|f\|_{L^2(\Omega)}$.

Proof of Lemma 7 (decay rate of $\text{osc}_{\mathcal{T}}(U)$)

- **Jump oscillation:** On an edge $S = T_1 \cap T_2$ with $j_S = \llbracket A \nabla U \rrbracket_S$, since ∇U is piecewise constant,

$$\begin{aligned} h_S^{1/2} \|j_S - \bar{j}_S\|_{L^2(S)} &\lesssim h_S^{1/2} \|A_1 - \bar{A}_1\|_{L^\infty(S)} \|\nabla U_1\|_{L^2(S)} \\ &\quad + h_S^{1/2} \|A_2 - \bar{A}_2\|_{L^\infty(S)} \|\nabla U_2\|_{L^2(S)} \lesssim \|A - \bar{A}\|_{L^\infty(\omega_S)} \|\nabla U\|_{L^2(\omega_S)}. \end{aligned}$$

- **Element oscillation:** Let

$$\text{osc}_{\mathcal{T}}(A, T) := h_T \|\text{div } A - \overline{\text{div } A}\|_{L^\infty(T)} + \|A - \bar{A}\|_{L^\infty(\omega_T)}.$$

- **Full oscillation:** We get

$$\text{osc}_{\mathcal{T}}(U, T) \lesssim \text{osc}_{\mathcal{T}}(f, T) + \text{osc}_{\mathcal{T}}(A, T) \|\nabla U\|_{L^2(\omega_T)}.$$

Proof of Lemma 7 (decay rate of $\text{osc}_{\mathcal{T}}(U)$)

- **Oscillation of A :** A scaling (Poincaré) argument gives

$$\text{osc}_{\mathcal{T}}(A, T) \lesssim h_T^{1+s_A-2/q} |A|_{W_q^{s_A}(T)}.$$

- **Greedy:** Running the greedy algorithm of Lecture 4 on $\text{osc}_{\mathcal{T}}(A, T)$ yields

$$\sup_{\mathcal{T} \in \mathbb{T}_N} \text{osc}_{\mathcal{T}}(A) \lesssim |A|_{W_q^{1+s_A}} N^{-(1+s_A)/2}.$$

- **Oscillation of f :** Applying Lemma 5 (decay rate of $\text{osc}_{\mathcal{T}}(f)$) gives

$$\sup_{\mathcal{T} \in \mathbb{T}_N} \text{osc}_{\mathcal{T}}(f) \lesssim |f|_{W_r^{s_f}} N^{-(1+s_f)/2}.$$

- **Overlay:** Overlaying the two near-best meshes for A and f , say $\mathcal{T} = \mathcal{T}_1 \oplus \mathcal{T}_2$, and using that $\#\mathcal{T} - \#\mathcal{T}_0 \leq 2N$ and $\|\nabla U\|_{L^2(\Omega)} \lesssim \|f\|_{L^2(\Omega)}$ gives the claim with $s = \min\{s_A, s_f\}$.

Membership in $\mathbb{A}_{1/2}$ with variable A

Theorem 2 (class $\mathbb{A}_{1/2}$ with variable A)

Let $d = 2$, polynomial degree 1, $p > 1$, $f \in L^2(\Omega)$, $A \in W_\infty^1(\Omega; \mathcal{T}_0)$, $u \in H_0^1(\Omega) \cap W_p^2(\Omega; \mathcal{T}_0)$ and Assumption 3 (initial labeling) be enforced. Then $(u, f, A) \in \mathbb{A}_{1/2}$ and

$$|(u, f, A)|_{1/2} \lesssim \|D^2 u\|_{L^p(\Omega; \mathcal{T}_0)} + \|f\|_{L^2(\Omega)} + \|A\|_{W_\infty^1(\Omega; \mathcal{T}_0)} \|f\|_{L^2(\Omega)}.$$

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$$|(u, f, A)|_{1/2} \lesssim \|D^2 u\|_{L^p(\Omega; \mathcal{T}_0)} + \|f\|_{L^2(\Omega)} + \|A\|_{W_\infty^1(\Omega; \mathcal{T}_0)} \|f\|_{L^2(\Omega)}.$$

Proof: We use the Greedy algorithm of Lecture 4.

- Greedy algorithm for $u \in W_p^2(\Omega)$ gives a mesh $\mathcal{T}_1$ with $\#\mathcal{T}_1 \leq N$ and

$$\|u - U\|_\Omega \lesssim \|D^2 u\|_{L^p(\Omega)} N^{-1/2}$$

- Lemma 7 (decay rate of $\text{osc}_{\mathcal{T}}(U)$) yields a mesh $\mathcal{T}_2$ with $\#\mathcal{T}_2 \leq N$ such that ($s_A = s_f = 1$)

$$\text{osc}_{\mathcal{T}_2}(U) \lesssim N^{-1/2} \left(f_{L^2(\Omega)} + \|A\|_{W_\infty^1(\Omega; \mathcal{T}_0)} \|f\|_{L^2(\Omega)} \right).$$

- Overlay $\mathcal{T}_1, \mathcal{T}_2$ and apply the definitions of σ_N and $\mathbb{A}_s$. □

ε -characterization of class $\mathbb{A}_s$ for general data

- **Exercise:** $(u, f, A) \in \mathbb{A}_s$ if and only if

for every $\varepsilon > 0$ there is a conforming refinement $\mathcal{T}_\varepsilon \geq \mathcal{T}_0$ and a function $V_\varepsilon \in \mathbb{V}(\mathcal{T}_\varepsilon)$ such that

$$E_{\mathcal{T}}(V_\varepsilon) \leq \varepsilon, \quad \#\mathcal{T}_\varepsilon - \#\mathcal{T}_0 \lesssim |(u, f, A)|_s^{1/s} \varepsilon^{-1/s}.$$

- **Role of $E_{\mathcal{T}}(V_\varepsilon)$:** The total error $E_{\mathcal{T}}(V_\varepsilon)$ is the analogue $\|u - U\|_\Omega$ in the earlier development for vanishing oscillation. This is the engine of cardinality bound below.
- **Interpretation:** Belonging to class $\mathbb{A}_s$ means that to achieve an error tolerance ε for $E_{\mathcal{T}}(V_\varepsilon)$ we need an order $\varepsilon^{-1/s}$ elements (or degrees of freedom or computational budget).

Quasi-optimal rates for general data

Dörfler marking is necessary (general data)

Lemma 8 (Dörfler marking: general data)

Let Assumption 1 (marking parameter) hold with the *stricter* threshold

$$\theta_* = \sqrt{\frac{C_2}{1+C_1(1+2C_0)}} \left(< \sqrt{\frac{C_2}{C_1}} \right),$$

C_0 be the Lipschitz constant of the estimator in its discrete argument, and $C_1 \geq C_2$ be the reliability/efficiency constants. If $\mu := \frac{1}{2}(1 - \theta^2/\theta_*^2) > 0$ and $\mathcal{T}_* \geq \mathcal{T}_k$ satisfies

$$\|u - U_*\|^2 + \text{osc}_{\mathcal{T}_*}^2(U_*) \leq \mu(\|u - U_k\|^2 + \text{osc}_{\mathcal{T}_k}^2(U_k)),$$

then the discrete neighborhood $\mathcal{R}^+$ of $\mathcal{R} = \mathcal{T}_k \setminus \mathcal{T}_*$ satisfies Dörfler marking

$$\mathcal{E}_{\mathcal{T}_k}(U_k, \mathcal{R}^+) \geq \theta \mathcal{E}_{\mathcal{T}_k}(U_k).$$

Proof — Steps 1–2

- **Step 1 — lower bound + hypothesis:** Using $C_2 \mathcal{E}_{\mathcal{T}_k}^2(U_k) \leq \|u - U_k\|^2 + \text{osc}_{\mathcal{T}_k}^2(U_k)$ yields

$$(1 - 2\mu)C_2 \mathcal{E}_{\mathcal{T}_k}^2(U_k) \leq (1 - 2\mu)(\|u - U_k\|^2 + \text{osc}_{\mathcal{T}_k}^2(U_k)).$$

Employing the hypothesis about total error reduction gives

$$(1 - 2\mu)C_2 \mathcal{E}_{\mathcal{T}_k}^2(U_k) \leq (\|u - U_k\|^2 - 2\|u - U_*\|^2) + (\text{osc}_{\mathcal{T}_k}^2 - 2\text{osc}_{\mathcal{T}_*}^2).$$

- **Step 2 — Pythagoras + localized upper bound:** Recalling Pythagoras

$$\|u - U\|^2 - \|u - U_*\|^2 = \|U - U_*\|^2,$$

and the localized discrete upper bound results in

$$\|u - U\|^2 - 2\|u - U_*\|^2 \leq \|U - U_*\|^2 \leq C_1 \mathcal{E}_{\mathcal{T}_k}^2(U_k, \mathcal{R}^+).$$

Proof — Steps 3–4

- **Step 3 — split $\mathcal{T}_k$:** Let $\mathcal{T}_k = \mathcal{R}^+ \dot{\cup} (\mathcal{T}_k \setminus \mathcal{R}^+)$. On $\mathcal{R}^+$, using $\text{osc}_{\mathcal{T}}(U, T) \leq \mathcal{E}_{\mathcal{T}}(U, T)$:

$$\text{osc}_{\mathcal{T}_k}^2(U_k, \mathcal{R}^+) - 2 \text{osc}_{\mathcal{T}_*}^2(U_*, \mathcal{R}^+) \leq \mathcal{E}_{\mathcal{T}_k}^2(U_k, \mathcal{R}^+).$$

On $\mathcal{T}_k \setminus \mathcal{R}^+ \subset \mathcal{T}_k \cap \mathcal{T}_*$, the Lipschitz continuity w.r.t the discrete argument of $\text{osc}_{\mathcal{T}_k}(U_k, T)$ for $T \in \mathcal{T}_k \cap \mathcal{T}_*$

$$|\text{osc}_{\mathcal{T}_k}(U_k, \mathcal{T}_k \setminus \mathcal{R}^+) - \text{osc}_{\mathcal{T}_*}(U_*, \mathcal{T}_k \setminus \mathcal{R}^+)| \leq C_0 \|U_k - U_*\|,$$

along with the localized upper bound yields

$$\text{osc}_{\mathcal{T}_k}^2(U_k, \mathcal{T}_k \setminus \mathcal{R}^+) - 2 \text{osc}_{\mathcal{T}_*}^2(U_*, \mathcal{T}_k \setminus \mathcal{R}^+) \leq 2C_0 \|U_k - U_*\|^2 \leq 2C_0 C_1 \mathcal{E}_{\mathcal{T}_k}^2(U_k, \mathcal{R}^+).$$

Adding these two estimates implies

$$\text{osc}_{\mathcal{T}_k}^2(U_k) - 2 \text{osc}_{\mathcal{T}_*}^2(U_*) \leq (1 + 2C_0 C_1) \mathcal{E}_{\mathcal{T}_k}^2(U_k, \mathcal{R}^+).$$

- **Step 4 — back into Step 1:** Combining Steps 2 and 3 give the Dörfler property in disguise

$$(1 - 2\mu)C_2 \mathcal{E}_{\mathcal{T}_k}^2(U_k) \leq (1 + C_1(1 + 2C_0)) \mathcal{E}_{\mathcal{T}_k}^2(U_k, \mathcal{R}^+). \quad \square$$

Cardinality of $\mathcal{M}_k$ (general data)

Lemma 9 (cardinality of $\mathcal{M}_k$: general data)

Let Assumptions 1 (marking parameter) with threshold θ_* as above, 2 (minimal marking) and 3 (initial labeling) hold. If $(u, f, A) \in \mathbb{A}_s$, $0 < s \leq \frac{1}{2}$ and $\#\mathcal{T}_k \geq 2\#\mathcal{T}_0$, then

$$\#\mathcal{M}_k \lesssim |(u, f, A)|_s^{1/s} \left(\|u - U_k\|_\Omega + \text{osc}_k(U_k) \right)^{-1/s} \quad \forall k \geq 0.$$

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$$\#\mathcal{M}_k \lesssim |(u, f, A)|_s^{1/s} (\|u - U_k\|_\Omega + \text{osc}_k(U_k))^{-1/s} \quad \forall k \geq 0.$$

- **Proof. Optimal mesh:** Set $\varepsilon^2 = \mu\Lambda_1^{-2}(\|u - U_k\|_\Omega^2 + \text{osc}_k^2)$. The ε -characterization gives $\mathcal{T}_\varepsilon$ with

$$E_{\mathcal{T}_\varepsilon}(V_\varepsilon)^2 \leq \varepsilon^2, \quad \#\mathcal{T}_\varepsilon \lesssim |(u, f, A)|_s^{1/s} \varepsilon^{-1/s}.$$

- **Overlay:** For $\mathcal{T}_* = \mathcal{T}_k \oplus \mathcal{T}_\varepsilon$, quasi-best approximation yields

$$\|u - U_*\|^2 + \text{osc}_{\mathcal{T}_*}^2(U_*) \leq \Lambda_1^2 E_{\mathcal{T}_\varepsilon}(V_\varepsilon)^2 \leq \Lambda_1^2 \varepsilon^2 = \mu(\|u - U_k\|^2 + \text{osc}_k^2).$$

- **Minimal cardinality:** Lemma 8 (Dörfler marking) applies, so the discrete neighborhood $\mathcal{R}^+$ of $\mathcal{R}$ satisfies Dörfler marking, whence minimal cardinality of $\mathcal{M}_k$ implies

$$\#\mathcal{M}_k \leq \#\mathcal{R}^+ \lesssim \mathcal{R} = \#\mathcal{T}_* - \#\mathcal{T}_k \lesssim \#\mathcal{T}_\varepsilon - \#\mathcal{T}_0 \lesssim |(u, f, A)|_s^{1/s} \varepsilon^{-1/s}. \quad \square$$

Quasi-optimality for general data

Theorem 3 (quasi-optimality for general data)

Let Assumptions 1 (marking parameter) with threshold θ_* as above, 2 (minimal cardinality), and 3 (initial labeling) hold. If $(u, f, A) \in \mathbb{A}_s$, $0 < s \leq \frac{1}{2}$, and $\#\mathcal{T}_k \geq 2\#\mathcal{T}_0$, then

$$\|u - U_k\|_{\Omega} + \text{osc}_k(U_k) \lesssim |(u, f, A)|_s (\#\mathcal{T}_k)^{-s} \quad \forall k \geq 1.$$

Proof:

• **Proof — Step 1:** Complexity of bisection and Lemma 9 (cardinality of $\mathcal{M}_k$)

$$\frac{1}{2}\#\mathcal{T}_k \leq \#\mathcal{T}_k - \#\mathcal{T}_0 \lesssim \sum_{j=0}^{k-1} \#\mathcal{M}_j \lesssim |(u, f, A)|_s^{1/s} \sum_{j=0}^{k-1} \left(\|u - U_j\|_{\Omega}^2 + \text{osc}_j^2 \right)^{-1/(2s)}.$$

Quasi-optimality — Steps 2–3

- **Step 2:** If γ is the scaling factor of the quasi-error in the contraction argument, the lower bound and $\text{osc}_j \leq \mathcal{E}_j$ give the equivalence

$$\|u - U_j\|_{\Omega}^2 + \gamma \text{osc}_j^2 \leq \|u - U_j\|_{\Omega}^2 + \gamma \mathcal{E}_j^2 \leq \left(1 + \frac{\gamma}{c_2}\right) (\|u - U_j\|_{\Omega}^2 + \text{osc}_j^2).$$

This implies

$$(\|u - U_j\|_{\Omega}^2 + \text{osc}_j^2)^{-1/(2s)} \lesssim (\|u - U_j\|_{\Omega}^2 + \gamma \mathcal{E}_j^2)^{-1/(2s)}.$$

- **Step 3:** Contraction of the quasi-error

$$\|u - U_k\|_{\Omega}^2 + \gamma \mathcal{E}_k^2 \leq \alpha^{2(k-j)} (\|u - U_j\|_{\Omega}^2 + \gamma \mathcal{E}_j^2) \quad j \leq k,$$

combined with convergence of the geometric series $\sum_j \alpha^{(k-j)/s}$ gives

$$\#\mathcal{T}_k \lesssim |(u, f, A)|_s^{1/s} (\|u - U_k\|_{\Omega}^2 + \text{osc}_k^2)^{-1/(2s)}.$$

Rearranging yields the asserted rate. □

Two corollaries with concrete rates

Corollary 2 (W_p^2 -regularity, piecewise-constant A)

Let $d = 2$, polynomial degree 1, $f \in L^2(\Omega)$, A piecewise constant, and $u \in W_p^2(\Omega; \mathcal{T}_0)$, $p > 1$. Then, for $\text{osc}_k(U_k) = \|h_k(f - \bar{f})\|_{L^2(\Omega)}$ and $\#\mathcal{T}_k \geq 2\#\mathcal{T}_0$, we have

$$\|u - U_k\|_{\Omega} + \text{osc}_k \lesssim (\|D^2 u\|_{L^p(\Omega; \mathcal{T}_0)} + \|f\|_{L^2(\Omega)}) (\#\mathcal{T}_k)^{-1/2}.$$

Corollary 3 (W_p^2 -regularity, variable A)

In addition to Corollary 3, let $A \in W_{\infty}^1(\Omega; \mathcal{T}_0)$. Then

$$\|u - U_k\|_{\Omega} + \text{osc}_k \lesssim (\|D^2 u\|_{L^p(\Omega; \mathcal{T}_0)} + \|f\|_{L^2(\Omega)} + \|A\|_{W_{\infty}^1(\Omega; \mathcal{T}_0)} \|f\|_{L^2(\Omega)}) (\#\mathcal{T}_k)^{-1/2}.$$

• **Proof:** Both proofs follow by combining the membership results with Theorem 3 (quasi-optimality).

Synthesis

Highlights of the general theory

- Outcomes:** The general theory removes two restrictive hypotheses of the vanishing-oscillation stage — **vanishing oscillation** and the **interior-node property** — at the price of:
 - a *stricter* threshold $\theta_* = \sqrt{C_2/(1 + C_1(1 + 2C_0))}$ ($< \sqrt{C_2/C_1}$);
 - measuring approximability of the triple (u, f, A) by the *total error* $E_{\mathcal{T}}(U) = \|u - U\|_{\Omega} + \text{osc}_{\mathcal{T}}(U)$;
 - driving the analysis by the *quasi-error* $\|u - U\|_{\Omega} + \mathcal{E}_{\mathcal{T}}(U)$.

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 - driving the analysis by the *quasi-error* $\|u - U\|_{\Omega} + \mathcal{E}_{\mathcal{T}}(U)$.

- Striking fact:**

A *single* Dörfler marking on the estimator still suffices — no separate marking of oscillation is required.

The complete picture

The three pillars of adaptivity combine cleanly in the general theory:

- ① **Reliability + efficiency** (Lecture 5): the quasi error is a faithful proxy for the total error.
- ② **Contraction of the total error** (Lecture 8): linear convergence.
- ③ **Optimal marking + complexity of bisection** (Lecture 9 and Lecture 4): the best-possible *rate*.

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- ➌ **Optimal marking + complexity of bisection** (Lecture 9 and Lecture 4): the best-possible *rate*.

AFEM achieves rate optimality with *no* a priori knowledge of the solution's singular structure — it discovers quasi-optimal meshes automatically from the computed indicators.

A new paradigm of AFEM

Beyond $A \in W_{\infty}^1, f \in L^2$

- **Standing assumptions:** The whole analysis so far assumed

$$A \in W_{\infty}^1(\Omega; \mathcal{T}_0), \quad f \in L^2(\Omega).$$

This entails **resolving A over $\mathcal{T}_0$** (no discontinuities of A inside elements of any refinement) and excludes forcing terms that are natural in applications.

- **The two data play different roles:**

The map $f \mapsto u$ is *linear* whereas the map $A \mapsto u$ is *nonlinear* (multiplicative).

- **Two ingredients circumvent these restrictions:**

- treat the forcing in $H^{-1}(\Omega)$;
- handle discontinuous A by *data approximation*.

They are combined into a **two-step AFEM**.

Forcing $f \in H^{-1}(\Omega)$: residual and localization

- **Natural loads:** $f \in L^p$ with $1 < p < 2$, or a line Dirac mass along a curve ($d = 2$). And H^{-1} is the right norm for the residual $R = f - \operatorname{div}(A \nabla U) \in H^{-1}(\Omega)$ because

$$M^{-1} \|R\|_{H^{-1}(\Omega)} \leq |u - U|_{H_0^1(\Omega)} \leq \alpha^{-1} \|R\|_{H^{-1}(\Omega)}.$$

- **Localization:** Galerkin orthogonality $\langle R, \phi_z \rangle = 0$ (interior nodes $z \in \mathcal{N}$) **localizes** the nonlocal H^{-1} -norm:

$$\frac{1}{1+d} \sum_{z \in \mathcal{N}} \|R\|_{H^{-1}(\omega_z)}^2 \leq \|R\|_{H^{-1}(\Omega)}^2 \leq (d+1) C_{\text{loc}} \sum_{z \in \mathcal{N}} \|R\|_{H^{-1}(\omega_z)}^2.$$

- **Goal:** approximate $\|R\|_{H^{-1}(\omega_z)}$ *without* replacing H^{-1} by a weighted L^2 -norm — the latter can *overestimate* (oscillation dominates the error, but not conversely): Bonito-N-DeVore, Kreuzer-Veeser.

Forcing $f \in H^{-1}(\Omega)$: an error-dominated estimator

- Projection onto discrete data:** For A piecewise constant and degree $n = 1$, $\operatorname{div}(A\nabla U)$ is a sum of Dirac masses on the skeleton with constant edge densities. Introduce a **projection** $\Pi_{\mathcal{T}} : H^{-1}(\Omega) \rightarrow \mathbb{F}_{\mathcal{T}}$ onto

$$\mathbb{F}_{\mathcal{T}} = \left\{ \ell \in H^{-1}(\Omega) : \langle \ell, w \rangle = \sum_S \int_S q_S w + \sum_{T \in \mathcal{T}} \int_T q_T w, \quad q_S, q_T \text{ const}, \quad w \in H_0^1(\Omega) \right\}.$$

- Redefined Galerkin solution:** Then $\Pi_{\mathcal{T}}(\operatorname{div}(A\nabla U)) = \operatorname{div}(A\nabla U)$, so $R - \Pi_{\mathcal{T}} R = f - \Pi_{\mathcal{T}} f$. Redefine the Galerkin solution U with $\Pi_{\mathcal{T}} f$ (**computable**: discrete data) instead of f . The residual $R_{\mathcal{T}} = \Pi_{\mathcal{T}} f + \operatorname{div}(A\nabla U) = -\Pi_{\mathcal{T}}(\operatorname{div}(A\nabla(u - U)))$ is computable and

$$\|R_{\mathcal{T}}\|_{H^{-1}(\Omega)} \lesssim |u - U|_{H_0^1(\Omega)}, \quad \operatorname{osc}_{\mathcal{T}}(f) = \|f - \Pi_{\mathcal{T}} f\|_{H^{-1}(\Omega)} \lesssim |u - U|_{H_0^1(\Omega)},$$

$$|u - U|_{H_0^1(\Omega)} \lesssim \|R_{\mathcal{T}}\|_{H^{-1}(\Omega)} + \operatorname{osc}_{\mathcal{T}}(f).$$

- Error-dominated estimator:** no overestimation — at the price of computing $\Pi_{\mathcal{T}} f$; possible for special classes of f (Kreuzer–Veese for $n = 1$, ActaNumer for any degree).

Discontinuous diffusion A : perturbation estimates

- **Naive estimate:** Approximate data $\widehat{\mathcal{D}} = (\widehat{A}, \widehat{f})$ of $\mathcal{D} = (A, f)$, with solutions $\widehat{u}, u$. The naive estimate

$$|u - \widehat{u}|_{H_0^1(\Omega)} \leq \widehat{\alpha} \left(\|f - \widehat{f}\|_{H^{-1}(\Omega)} + \alpha^{-1} \|A - \widehat{A}\|_{L^\infty(\Omega)} \|f\|_{H^{-1}(\Omega)} \right)$$

is problematic for discontinuous A (it needs the jumps of A to be matched exactly).

- **Perturbation estimate:**

$$|u - \widehat{u}|_{H_0^1(\Omega)} \leq \widehat{\alpha} \left(\|f - \widehat{f}\|_{H^{-1}(\Omega)} + \|\nabla u\|_{L^p(\Omega)} \|A - \widehat{A}\|_{L^q(\Omega)} \right), \quad q = \frac{2p}{p-2},$$

provided $u \in W_p^1(\Omega)$, $p \geq 2$ ($p = 2$ recovers the naive one).

- **Caveats:**

- Needs $\widehat{A}$ *uniformly* positive definite and $u \in W_p^1$ (Meyers: guaranteed if $f \in W_p^{-1}$, even for $A \in L^\infty$).
- Approximating A in L^q under uniform ellipticity is delicate — trivial for $n = 0$ via elementwise means.

Two-step AFEM (AFEM-TS)

- **Idea:** Approximate the data first, then solve — the algorithm of Stevenson / ActaNumer.

AFEM-TS ($\mathcal{T}_0, \varepsilon_0, \omega$)

For $\mathcal{T}_0$ satisfying Assumption 3 (initial-labeling) and $0 < \omega \ll 1$, let $k = 0$; loop:

$$[\widehat{\mathcal{T}}_k, \widehat{\mathcal{D}}_k] = \text{DATA}(\mathcal{T}_k, \mathcal{D}_k, \omega \varepsilon_k)$$

$$[\mathcal{T}_{k+1}, U_{k+1}] = \text{GALERKIN}(\widehat{\mathcal{T}}_k, \widehat{\mathcal{D}}_k, \varepsilon_k)$$

$$\varepsilon_{k+1} = \frac{1}{2} \varepsilon_k; \quad k \leftarrow k + 1$$

- **DATA:** approximates $(A, f) \rightarrow (\widehat{A}, \widehat{f})$ to accuracy $\tau = \omega \varepsilon \ll \varepsilon$ (greedy-like), producing a conforming refinement $\widehat{\mathcal{T}} \geq \mathcal{T}$.
- **GALERKIN:** refines $\widehat{\mathcal{T}}$ to a conforming $\mathcal{T}$ with $\mathcal{E}_{\mathcal{T}}(U_{\mathcal{T}}) \leq \varepsilon$.
- **Discrete data:** If the data is already discrete ($\mathcal{D} = \widehat{\mathcal{D}}$), DATA is skipped and AFEM-TS reduces to the one-step GALERKIN.

Analysis of Two-step AFEM (AFEM-TS)

- **GALERKIN:** Its analysis is similar to Lectures 8 and 9 for pw constant A .
- **DATA:** This uses a greedy algorithm as in Lecture 4.
- **Convergence:** AFEM-TS converges provided ω is sufficiently small, namely that data (f, A) are approximated a bit better than the solution u – to a comparable accuracy.
- **Complexity:** It results from combining the optimal complexities of DATA and GALERKIN, very much in the spirit of last discussion of Lecture 9.

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THANK YOU !